

TS Question Paper 2018**Class 12 Chemistry****Paper - II****Time:3 Hours****Max.Marks:60**

Note: Read the following instructions carefully:

1) Answer all the questions of Section-‘A’. Answer any six questions from Section-‘B’ and any two questions from Section-‘C’.

2) In Section-‘A’, Questions from Sr. Nos. 1 to 10 are of “very short answer type”. Each question carries two marks. Every answer may be limited to two to three sentences. Answer all the questions at one place in the same order.

3) In Section-‘B’, Questions from Sr. Nos. 11 to 18 are of “Short answer type”. Each question carries four marks. Every answer may be limited to 75 words.

4) In Section-‘C’, Questions from Sr.Nos.19 to 21 are of “Long answer type”. Each question carries eight marks. Every answer may be limited to 300 words.

5) Draw labelled diagram, wherever necessary for questions in section-‘B’ and ‘C’.

SECTION- A **$10 \times 2 = 20$** **Note: Answer all the questions:****1. State Raoult's law.**

Ans: Raoult's law : The partial vapour pressure of any volatile component of a solution is the product of vapour pressure of that pure component and the mole fraction of the component in the solution.

$$P = p_O X$$

Where p is the vapour pressure of a component of the solution, p_O is the vapour pressure of pure component and X is the mole fraction of that component in the solution.

2. State Faraday's second law of electrolysis.

Ans: Faraday's second law of electrolysis states that When the same quantity of electricity is passed through different electrolytes, the masses of different ions that are liberated at the electrodes are directly proportional to their chemical equivalent weights.

$$m \propto E$$

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$

3. What is PHBV? How is it useful to man?

Ans: PHBV is a copolymer. It is polyhydroxy butyrate -CO- β -hydroxy valerate. The monomers used are 3-hydroxy butanoic acid and 3- hydroxy pentatonic acid. It has an ester linkage. It is useful to mankind because it is biodegradable polymer and thus prevents environmental pollution.

4. Write the names of the monomers for the following polymers.

a) Bakelite b) Nylon 6, 6

Ans: a) Phenol, Formaldehyde

b) Adipoyl chloride and hexamethylene diamine

5. What is a ligand? Give one example for unidentate ligand.

Ans: Ligand may contain one or more than one donor atoms. If only one donor atom is present in its molecule which can coordinate, then it is unidentate. For example, H_2O , NH_3 .

If the ligand contains two donor atoms placed in a way that a five or six membered ring is formed with the metal ion, then it is called as bidentate. Example is oxalate ion.

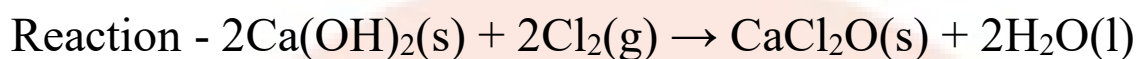
6. What are the neutral oxides of nitrogen?

Ans: Neutral oxides are those oxides which show neither basic nor acidic properties when they react with water. Examples include nitrous oxide (N_2O) which are only slightly soluble in water, and nitric oxide (NO) which is appreciably soluble in

cold water.

7. What happens when Cl_2 reacts with dry slaked lime?

Ans: When dry or slaked lime reacts with chlorine gas then it forms mostly calcium hypochlorite, calcium chloride with water.



8. What is blister copper? Why is it so called?

Ans: Definition of blister copper: metallic copper of a black blistered surface, being the product of converting copper matte and being about 98.5 to 99.5 per cent pure. — called also Bessemer copper.

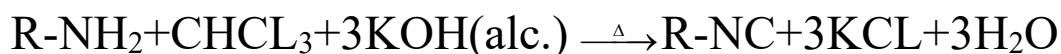
An almost pure copper produced in an intermediate stage of copper refining is called blister copper.

It is called so because in impure form copper has a blister-surface due to release of gas during cooling.

9. Explain carbylamines reaction with an example.

Ans: Primary amine (both aliphatic and aromatic) when warmed with chloroform and alcoholic KOH, gives isocyanides (carbylamines). This is called carbylamines reaction.

Carbylamines has an offensive smell. This reaction is answered only by primary amine and hence to distinguish primary amine from other classes of amines.



Primary chloroform Potassium Carbylamines

amine

hydroxide

10. Explain the reaction of Aniline with Nitrous acid.

Ans: when the aniline is react with the nitrous acid it form the diazonium salt as a product. This reaction can occur in two ways

1. it reacts with nitrous acid under cold condition.
2. it reacts with nitrous under in room temperature

Reaction under cold condition,

It gives the diazonium chloride as a by product of aniline and nitrous acid.

Reaction under in room temperature,

It gives the phenol as a by product of aniline and nitrous acid reaction.

SECTION- B**(6×4=24)**

Note: Answer Any Six questions.

11. Explain Schottky and Frenkel defects.

Ans: Schottky defect occurs when oppositely charged atoms (cation and anion) leave their corresponding lattice sites and create a pair of Vacancy Defects. Since both cation and anion leave the lattice sites at the same time, so overall electrical neutrality of the crystal is maintained; however, density reduces because of the vacancies. Schottky defects occur in ionic crystals where the size of anion is almost same with the size of the cation.

Frenkel Defect is one type of point defect where an atom (better to say ion, especially cation) leaves its original lattice site and occupies an interstitial position on the same crystal. Usually, this type of defect is observed in ionic solids, where size of anion is substantially larger than the size of cation.

12. Define mole fraction. Calculate the mole fraction of H_2SO_4 in a solution containing 98%(w/w) H_2SO_4 by mass.

Ans: Mole fraction is a unit of concentration, defined to be equal to the number of moles of a component divided by the total number of moles of a solution. Because it is a ratio, mole fraction is a unitless expression. The mole fraction of all components of a solution, when added together, will equal 1. 98% aqueous H_2SO_4 solution but mass contains 98g of H_2SO_4

and 2g of water.

No. of moles of $\text{H}_2\text{SO}_4 = (98\text{g})/(98\text{g/mol}) = 1 \text{ mol}$,

No = $(2\text{g}/18\text{g/mol}) = 0.11 \text{ mol}$

No. of moles of H_2 mole fraction = No. of moles of solute/ total no. of moles in solution

Mole fraction of $\text{H}_2\text{SO}_4 = 1/1.11 = 0.9$

13. What are emulsions? How are they classified? Give one example for each.

Ans:

Emulsion: The colloidal solution in which both the dispersed phase and dispersion medium are liquids is called an emulsion.

Emulsions are classified as-

Oil in water type- The emulsion in which oil is present as the dispersed phase and water as the dispersion medium (continuous phase) is called an oil-in-water emulsion

E.g.- Milk is an example of the oil-in-water type of emulsion.

Water in oil type- The emulsion in which water forms the dispersed phase, and the oil acts as the dispersion medium is called a water-in-oil emulsion.

E.g.- Butter is an example of the water-in-oil type of emulsion.

14. Explain calcinations and roasting with suitable examples.

Ans: For example, carbonate ores are converted into the oxide by heating in the absence of air. eg: $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$.
Roasting: The sulphide ores are converted into oxides by heating strongly in the presence of excess of air. ... Calcination: The carbonate ores are changed into oxides by heating strongly in limited air

15. Explain Werner's theory coordination compounds with suitable examples.

Ans: Werner's theory coordination compounds:

- Two types of valency are exhibited by the central metal atom. Example- primary valency and the secondary valency.
- Secondary valency are fixed in number in the every central metal atom. Hence it has a fixed coordinate number.
- Both primary and secondary valences are satisfied by the metal atom, in which primary valency is satisfied by negative ion and whereas secondary valency are satisfied with neutral molecules.
- Definite geometry of coordinated compound is caused by the secondary valency which is always directed towards fixed position in space.

16. What are hormones? Give one example for each of the following:

- a) Steroid hormones
- b) Polypeptide hormones
- c) Amino acid derivatives

Ans: A hormone is any member of a class of signaling molecules, produced by glands in multicellular organisms, that are transported by the circulatory system to target distant organs to regulate physiology and behavior.

- a) Sex hormones (androgens, estrogens, progesterone)
- b) Insulin, Prolactin
- c) Epinephrine and norepinephrine

17. What are analgesics? How are they classified? Give one example for each.

Ans: Analgesics may be classified into two types: anti-inflammatory drugs, which alleviate pain by reducing local inflammatory responses; and the opioids, which act on the brain. The opioid analgesics were once called narcotic drugs because they can induce sleep.

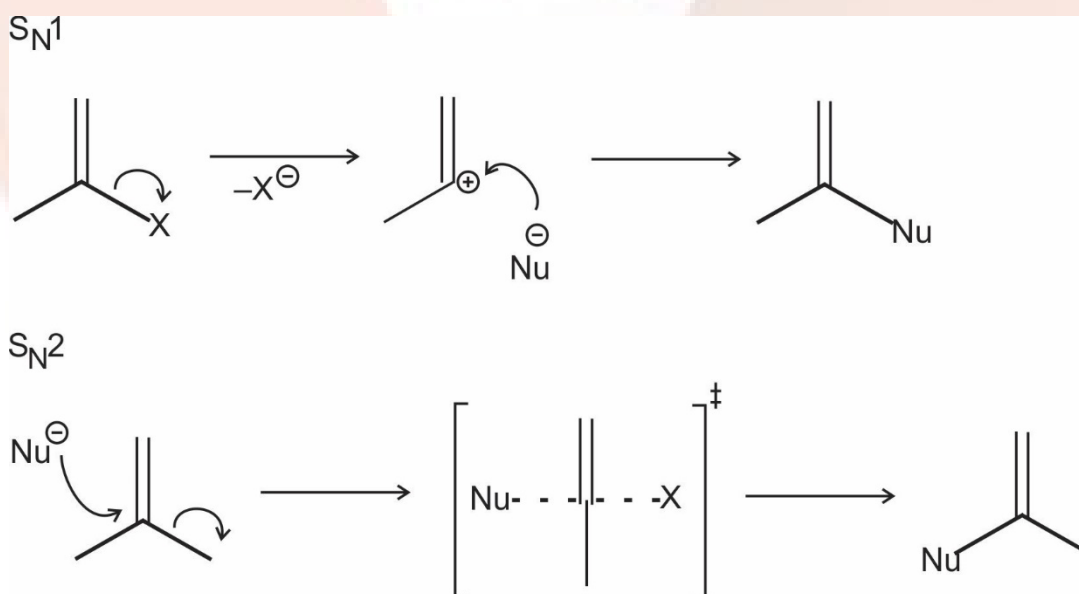
The medicines which abolish or reduce pain causing impairment of consciousness, mental confusion or some other nervous system disturbance.

Ex:-Paracetamol

18. Explain S_N1 and S_N2 reactions with an example.

Ans: S_N1 reactions are unimolecular in rate of reaction and have a stepwise mechanism. This process first involves bond cleavage by the LG to generate a carbocation intermediate. The stability of carbocation formation will determine if S_N1 or S_N2 reactions occur.

S_N2 reactions are bimolecular in rate of reaction and have a concerted mechanism. The process involves simultaneous bond formation by the nucleophile and bond cleavage by the leaving group. The transition state looks like this. Because the reaction is concerted, S_N2 mechanisms will always lead to an inversion of stereochemistry!



SECTION – C

(2 × 18 = 16)

Note : Answer ANY TWO questions.

19. a) What are Galvanic cells? Explain the working of a Galvanic cell by taking Daniel cell as an example.

Ans: Galvanic cells – An electrical energy from redox reaction takes place within the cell via electrochemical cell. This cell is known as galvanic cell.

Working of a galvanic cell:

- It involves a chemical reaction which makes the electric energy.
- In the redox reaction, galvanic cell uses the energy transfer between the electrons which converts the chemical energy into electric energy.
- Galvanic cell is used to separate the flow of electrons. This flow of electrons is basically known as current.
- This galvanic cell consists of two metals, these two metals form the anode and other form the cathode.

Example : two metals involved in chemical reaction in which zinc ends up losing two electrons, and these electrons are taken by the copper. If these two metals are placed in separate containers they would be connected by the conducting wire, an electric current is generated in them. In this case, two solutions are not mixed directly but can be joined by the salt bridge.

b) Describe the salient features of the collision theory of reaction rates.

Ans: Collision features of reaction rates:

- According to the theory of collision, a chemical reaction takes place only by collision between the reacting molecules, but all the collisions are not effective.
- The colliding molecules must possess enough kinetic energy to cause a reaction.
- Reacting molecules must collide with proper orientation.
- Only a small fraction of the molecule collides effectively to give the product.
- Two main condition are there for a collision between the reacting molecules to be productive or effective.

20.a) Write the important reactions involved in the manufacture of sulphuric acid by ‘contact forces’.

Ans: $S(s) + O_2(g) \rightarrow SO_2(g)$

$4FeS_2(s) + 11O_2(g) \rightarrow 2Fe_2O_3(s) + 8SO_2(g)$

$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g) \Delta H 196 \text{ KJ mol}^{-1}$

$H_2SO_4(l) + SO_3(g) \rightarrow H_2S_2O_7$

$H_2S_2O_7(l) + H_2O \rightarrow 2H_2SO_4(l)$

b) Write the prepared of the comounds XeF_2 and XeF_4 . Give their structures.

Ans: 1. Xenon difluoride (XeF_2)

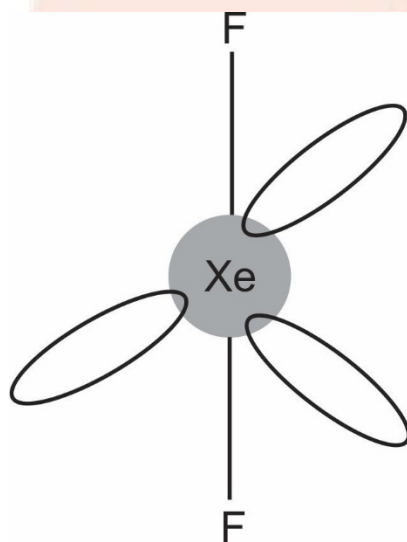
It is prepared by heating a mixture of Xenon and fluorine in the ratio 2:1 at 400 degree Celsius and 1 bar pressure in a sealed nickel tube.



XeF_2 undergoes hydrolysis when treated with water and evolves oxygen.

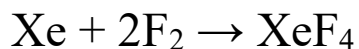


In XeF_2 , Xenon is sp_3d hybridised and the molecule has linear structure as shown.



2. Xenon tetrafluoride (XeF_4)

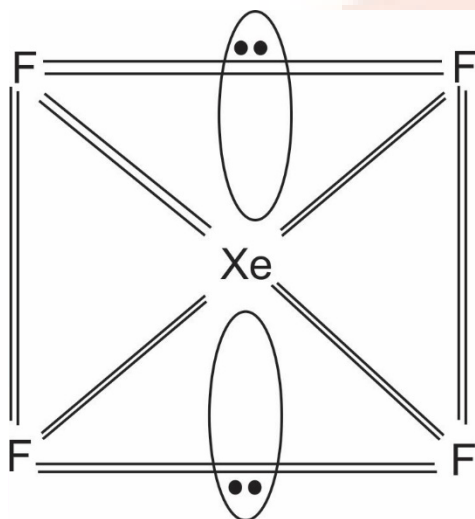
It is prepared by heating a mixture of Xe and F_2 in the molecular ratio 1:5 at 400 degree Celsius and 6 atm in a sealed nickel tube.



XeF_4 react with water and produces explosive XeO_3



In XeF_4 , Xenon is in sp_3d_2 hybridised state and has square planar geometry.



XeF_4

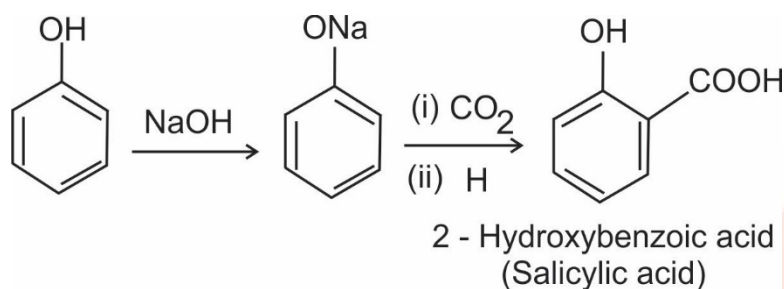
21. Explain the following reactions with equations.

a) Kolbe's reaction

Ans: When phenol is treated with sodium hydroxide, phenoxide ion is generated. The phenoxide ion generated is more reactive than phenol towards electrophilic aromatic substitution reaction. Hence it undergoes electrophilic substitution reaction with carbon dioxide which is a weak electrophile. Ortho-hydroxybenzoic acid (salicylic acid) is formed as the major product.

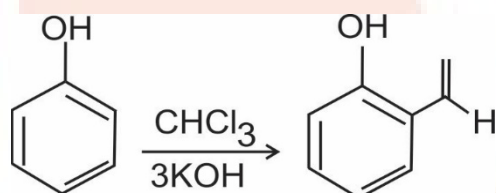
This reaction is popularly known as Kolbe's reaction.

It is named after the scientist Hermann Kolbe.



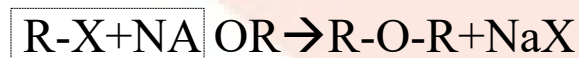
b) Reimer-Tiemann-reaction

Ans: When phenols are treated with chloroform in the presence of sodium hydroxide, an aldehyde group (-CHO) is introduced at the ortho position of benzene ring leading to the formation of o-hydroxybenzaldehyde. This reaction is popularly known as Reimer Tiemann reaction.



c) Williamson's ether synthesis

Ans: Action of alcoholic sodium alkoxide on alkyl halides to form ethers is known as Williamson's synthesis.



Sodium alkoxide Ether

d) HVZ reaction

Ans: The Hell–Volhard–Zelinsky halogenation reaction halogenates carboxylic acids at the α carbon. The reaction is

named after three chemists, the German chemists Carl Magnus von Hell and Jacob Volhard and the Russian chemist Nikolay Zelinsky

